PRAYCG Formula Module Report

v1.2 - Integrated Edition

Single current specification
This edition reorganizes the complete formula catalog into one current document. Superseded wording has been reconciled in place; there is no historical core followed by a revision appendix.

Scope	Current version
Analysis suite	Master Comprehensive Analysis Suite v1.6.1
Control Center	PRAYCG Control Center v0.95
Protocol library	PRAYCG3 v3.0, PRAYCG4 v4.0, Semantic Meaning Gradient v1.0
Approved author	Hoyt Banks
Contact	hoytbanks@gmail.com
Public repository	github.com/hbanks87/praycg-open
OSF project	osf.io/8n75v
Document license	CC BY 4.0 unless a file states otherwise

Central boundary. These formulas define operational indices, analysis gates, controls, and development candidates. They do not prove consciousness, narrative reception, memory formation, clinical state, cellular hidden variables, microtubules, biophotons, morality, or literal thermodynamic energy flow.

Prepared as a public-facing technical specification with explicit evidence grades, missingness rules, provenance requirements, and falsification boundaries.

Contents

PART I
Scope, evidence, and notation

Defines what the suite can claim, how quantities are normalized, and how unavailable evidence is handled.

1. Purpose and scientific boundary

GOVERNING INTERPRETATION

This report is the current formula catalog for the PRAYCG runner, Master Comprehensive Analysis Suite, and associated exploratory modules. It makes each computation auditable while keeping software implementation separate from construct validity.

Tier	Role
Primary candidate path	Timing/QC + stimulus provenance + artifact/confound gates -> A-MRED or MRED-Peak/Resolution at locked anchors.
Secondary summaries	NIP, BIT, CII, IAQ, TTI, NUPI, DGA, and baseline summaries when declared by the study.
Exploratory isolated	CAI/SID, Micro Handoff, continuous autonomic coupling, KHT-topo, NAST, EET, MRED-ITP, and other development candidates.
Attribution defenses	Phase, ShotOrder, Override, HOC-R, OSA, OHC, AAM, RespDualPath, CET-R, artifacts, and provenance.

Epistemic rule
A formula may be implemented correctly while the intended construct remains unvalidated. Report both software status and evidence status.

2. Conditions, streams, and common transforms

SHARED NOTATION

```
B1 = Baseline 1
P = PhaseScrambled control
S = ShotOrderScramble control
T = intact Target watched naturally
O = Contextual Override under analytic task stance
B2 = Baseline 2 / final reflection

x_EEG,ch(t) = EEG channel stream
RR(t) = cardiac beat intervals
resp(t) = respiration stream
u(t) = exogenous stimulus vector
m(t) = marker stream
s(t) = self-report metadata

z_B1(x_t) = 0.67448975 * (x_t - median_B1(x)) / MAD_B1(x)
MAD_B1(x) = median_B1(|x_t - median_B1(x)|)
sigmoid(x) = 1 / (1 + exp(-x))
[x]_+ = max(x,0)
clip01(x) = min(1,max(0,x))
```

The robust transform is algebraically equivalent to dividing by 1.4826 times MAD. Every module must declare its reference distribution. Micro Handoff requires valid Baseline 1 and permits no full-run fallback.

3. Evidence grades, missingness, and authority

REQUIRED REPORTING RULE

State	Meaning
PASS	Declared computational requirements were satisfied.
CAUTION	Computation is possible, but limitations must travel with the result.
INELIGIBLE	The endpoint must not be interpreted for that run.
NOT_APPLICABLE	The protocol lacks the required condition or input.
FAILED	Execution failed; downstream absence is not evidence.
MISSING	A required quantity is unavailable; it is not zero or favorable evidence.

Authority order

Use the versioned implementation and frozen configuration first, then the protocol/formula specification, then provenance/QC/warnings, and finally explanatory theory documents.

PART II
Protocols, media, and experimental controls

Defines what is shown, how structural controls differ, and how every media derivative is traced.

4. PRAYCG3 v3.0

IMPLEMENTED FIXED-ORDER PROTOCOL

```
BASELINE_1
 -> PHASE_SCRAMBLED -> washout/report
 -> TARGET -> washout/report
 -> CONTEXTUAL_OVERRIDE -> washout/report
 -> BASELINE_2_REFLECTION -> final report
```

Baseline 1, all three washout/report periods, Baseline 2/final reflection, and the final report are retained. Target and Override use the same cue-embedded media; only task stance changes.

```
Delta_TP = mean(Y_Target) - mean(Y_PhaseScrambled)
Delta_TO = mean(Y_Target) - mean(Y_Override)
```

Fixed-order caution
A contrast from one fixed order combines condition, period, and carryover. Historical PRAYCG3 results remain exploratory unless order is defensibly modeled or prospectively counterbalanced.

5. PRAYCG4 v4.0 and ShotOrder

IMPLEMENTED FIXED-ORDER PROTOCOL

```
BASELINE_1
 -> PHASE_SCRAMBLED -> washout/report
 -> SHOT_ORDER_SCRAMBLE -> washout/report
 -> TARGET -> washout/report
 -> CONTEXTUAL_OVERRIDE -> washout/report
 -> BASELINE_2_REFLECTION -> final report
```

```
V_Target = S_1 (+) S_2 (+) ... (+) S_m
V_ShotOrder = S_pi(1) (+) S_pi(2) (+) ... (+) S_pi(m)
```

The permutation changes cross-shot order while retaining local shots, faces, bodies, objects, voices, and biological motion as far as the editing pipeline permits. The manifest records source and derivative hashes, shot boundaries, permutation, and settings.

```
Delta_TP = mean(Y_T) - mean(Y_P)
Delta_TS = mean(Y_T) - mean(Y_ShotOrder)
Delta_TO = mean(Y_T) - mean(Y_O)
```

Contrast	Question	Caution
Target - Phase	Intact target versus strongly structure-damaged input	Phase scrambling also changes local contours, faces, objects, and motion.

Contrast	Question	Caution
Target - ShotOrder	Coherent temporal sequence beyond locally intact shots	New cuts, audio discontinuities, order, and familiarity remain.
Target - Override	Natural viewing versus analytic extraction	Task load, gaze, cue monitoring, and task relief remain.

ShotOrder narrows the structural question but does not uniquely isolate narrative meaning.

6. Semantic Meaning Gradient v1.0

CONDITIONAL ACQUISITION MODULE

```
SEMANTIC_ZERO_1 -> HIGH_MEANING_TARGET_1 -> ARITHMETIC_OVERRIDE_1
```

Semantic-Zero must be a distinct coherent low-meaning stimulus. PhaseScrambled or chaotic media is not substituted automatically because randomness may create novelty, threat, prediction error, or puzzle demand.

```
SMD(t) = sigmoid[w_H H + w_A A + w_V V + w_C C + w_R R + w_K K + w_T T
                 - w_P Puzzle - w_O Overload]
```

H, A, V, C, R, K, and T denote human/social presence, agency, vulnerability, causal continuity, repair/care, stakes, and transformation. Weights must be nonnegative, normalized, and frozen independently of test physiology.

```
M_total = (1/T) integral_0^T SMD(t) dt
M_peak = max_t SMD(t)
M_arc_star = a Var[SMD(t)] + b T0 max_t |dSMD(t)/dt|
```

```
SZI = (1-Meaning)(1-Empathy)(1-Absorption)(1-Threat)(1-PuzzleDemand)
OPI = TaskCompliance * AnalyticEffort * (1-Empathy)(1-Meaning)(1-NarrativeEngagement)
DeltaM = M_target - M_semantic_zero
MCS = mean(StoryActive, EmotionalAfterglow, Replay, FeltReturnDelay)
```

> Validation boundary
> SZI, OPI, DeltaM, MCS, and MRI are proposed concepts. Inputs must be explicitly mapped to [0,1]. The current runner records but does not compute MRI.

7. Prospective PRAYCG3/PRAYCG4 sequence package

PROSPECTIVE TEMPLATE

Separately named prospective variants use Williams first-order-balanced sequences and lock the assigned runner before acquisition. The public fixed-order runners remain available for historical comparability.

```
Period balance: N_(period,condition) approximately constant
Carryover balance: N_(a->b) approximately constant for ordered pairs a != b
j(p,s) = ((p-1) + (s-1)) mod m
```

Variant family	Orders
PRAYCG3	Six sequences covering Phase, Target, and Override with reversed balance.
PRAYCG4	Four sequences covering Phase, ShotOrder, Target, and Override.

```
h_f = SHA256(bytes(f))
```

Every critical runner, configuration, stimulus, derivative, anchor, and manifest is hashed before collection. Recognition, comprehension, confidence, narrative coherence, and subjective outcomes remain independent criteria rather than CAI ingredients.

8. MediaPrep formulas

IMPLEMENTED STIMULUS PREPARATION

MediaPrep prepares the stimulus suite. It does not certify physiology.

8.1 Cue schedule

For fixed-interval number cues:

```
t_i_start = t_start_delay + (i - 1) * T_cue_interval
t_i_end = t_i_start + T_display
```

where typical PRAYCG settings are:

```
T_cue_interval = 3.0 s
T_display = 0.85 s
cue_position = upper_right
```

The running sum is:

```
S_i = S_{i-1} + v_i
S_0 = 0
```

where v_i is the cue value.

8.2 Target/Override identity rule

The Target and Override media are generated from the same cue-embedded stimulus:

```
hash(Target_video) = hash(Override_video)
```

The only intended difference is participant instruction:

Target instruction: watch naturally.

Override instruction: perform analytic cue task.

8.3 Phase-scrambled Control

A simplified visual phase-scramble operation can be written in Fourier form:

```
F_target(k) = A(k) * exp(i * phi(k))
F_control(k) = A(k) * exp(i * phi_random(k))
control_frame = inverse_Fourier(F_control)
```

The intent is:

preserve low-level audiovisual structure remove recognizable narrative meaning The Control is generated from the cue-embedded Target so the cue rhythm and cue energy remain represented.

8.4 ALS / photodiode stimulus pulse

The ALS timing square is an exogenous display-timing regressor:

```
ALS_pulse(t) = 1 if t in [video_start, video_start + pulse_duration]
ALS_pulse(t) = 0 otherwise
```

Pass criteria include:

visible analog rise return to baseline no clipping for full pulse duration pulse present for Control, Target, and Override Boundary:

ALS_pulse(t) belongs to u(t), not Y_topo(t).

9. PhaseScrambled control as sensory-entrainment guard

REQUIRED STRUCTURAL CONTROL

9.1 Why this update exists

The phase-scrambled branch should not be described as physiologically meaningless. It is a meaning-damaged sensory-control condition. It can still drive visual entrainment, auditory entrainment, boredom, confusion, artifact, task-like scanning, startle, or low-level affect. Its job is not to be dead. Its job is to preserve enough low-level stimulus structure that Target-specific effects must survive a credible sensory-entrainment alternative.

The updated control definition is:

```
TargetVideo = cue_embed(clean_master)
OverrideVideo = TargetVideo
ControlVideo = phase_scramble(TargetVideo)
ControlAudio = speech_shaped_noise_envelope(TargetAudio)
```

The control branch can be modeled as:

```
Control(t) = SensoryDrive(t) + CueEnergy(t) + ArtifactRisk(t) - RecognizableNarrative(t)
```

Thus the serious endpoint is not:

```
Control = 0
```

but:

```
Target > Control
```

after artifact, stimulus-fingerprint, and CET-R gates.

9.2 Formal sensory-entrainment model

Let the exogenous stimulus vector be:

```
u(t) = [audio_env(t), luminance(t), visual_change(t), optical_flow(t), cut_train(t),
cue_train(t), ALS_pulse(t)]
```

A sensory-tracking prediction for a physiological feature is:

```
Y_feature(t) = beta_0 + sum_m beta_m u_m(t - tau_m) + beta_art Artifact(t) + epsilon(t)
```

The residualized feature is:

```
Y_resid(t) = epsilon(t)
```

A Target-specific meaning candidate must remain after this operation:

```
MeaningCandidate(t) = f(Y_resid(t), D_gate(t), MR(t), ENC(t))
```

9.3 Control validity gates

The phase-scrambled branch remains required for serious PRAYCG. It can be skipped only for engineering tests or PRAYCG-lite demonstrations. Confirmation-grade interpretation requires:

```
ControlAudioQC = PASS
ControlVisualQC = PASS
ControlGeneratedFromCueEmbeddedTarget = TRUE
TargetOverrideHashMatch = TRUE
CET_R_available = TRUE
```

If Control is absent:

```
claim_grade = PRAYCG_LITE_NO_SENSORY_CONTROL
```

If Control audio remains recognizable:

```
claim_grade = PILOT_ONLY_CONTROL_AUDIO_FAIL
```

10. StimulusFingerprint v1.8

IMPLEMENTED STIMULUS CHARACTERIZATION

StimulusFingerprint estimates the physical stimulus structure so physiological modules can be residualized against exogenous drive.

10.1 Visual luminance

For frame f with pixels P_f(x,y) converted to grayscale intensity:

```
L_f = mean_{x,y}(gray(P_f(x,y)))
```

Resampled to time:

```
L(t) = interpolate({frame_time_f, L_f})
```

10.2 Visual change / motion proxy

A simple frame-difference proxy:

```
Vchange_f = mean_{x,y}( |gray(P_f) - gray(P_{f-1})| )
```

An optical-flow proxy may be substituted:

```
Flow_f = mean_{x,y}( sqrt(u_flow(x,y)^2 + v_flow(x,y)^2) )
```

10.3 Histogram delta

```
H_f = normalized_histogram(gray(P_f))
HistDelta_f = ||H_f - H_{f-1}||_1
```

10.4 Cut train

```
Cut_f = 1[HistDelta_f > theta_cut]
```

where theta_cut is calibrated from the distribution of frame changes.

10.5 Flash-event proxy

```
Flash_f = 1[ |L_f - L_{f-1}| > theta_flash ]
```

10.6 Audio RMS envelope

For audio waveform a[n], sample rate fs, and window length N:

```
RMS_k = sqrt( (1/N) * sum_{n in window k} a[n]^2 )
dBFS_k = 20 * log10(RMS_k + eps)
```

Envelope derivative:

```
AudioDeriv_k = RMS_k - RMS_{k-1}
```

10.7 Cue regressors

```
CueOn(t) = 1 if any cue visible at t, else 0
CueImpulse(t) = sum_i delta(t - t_i_start)
CueValue(t) = v_i during cue i, else 0
CuePhaseSin(t) = sin(2*pi*t / T_cue_interval)
CuePhaseCos(t) = cos(2*pi*t / T_cue_interval)
```

The cue frequency is:

```
f_cue = 1 / T_cue_interval
```

For T_cue_interval = 3.0 s:

```
f_cue = 0.333 Hz
```

10.8 Dominant frequency map

For each stimulus regressor u_m(t), compute a short-time Fourier transform:

```
U_m(f,tau) = STFT[u_m(t)]
```

Dominant local frequency:

```
f_m_star(tau) = argmax_f |U_m(f,tau)|^2
```

10.9 Anchor stimulus vector

For anchor window W_j:

```
StimVec_j = [mean(L), std(L), mean(Vchange), cut_rate, flash_rate,
```

mean(AudioRMS), std(AudioRMS), cue_density, f_star_audio, f_star_visual]_Wj This vector feeds CET/EET and stimulus-side QC.

PART III
Acquisition, signal features, and quality control

Builds the timebase, EEG and autonomic features, artifact gates, and hardware timing evidence.

11. Data ingestion and timebase reconstruction

IMPLEMENTED INFRASTRUCTURE

11.1 Marker-defined phase slicing

The runner writes marker pairs such as:

TARGET_1_START, TARGET_1_END CONTROL_1_START, CONTROL_1_END CONTEXTUAL_OVERRIDE_1_START, CONTEXTUAL_OVERRIDE_1_END BASELINE_1_START, BASELINE_1_END BASELINE_2_REFLECTION_START, BASELINE_2_REFLECTION_END For each phase p:

```
phase_interval_p = [m_start,p, m_end,p]
```

A sample belongs to phase p if:

```
m_start,p <= t_sample < m_end,p
```

11.2 Continuous-stream reconstruction

If raw LSL timestamps are irregular, the suite can reconstruct time from status markers or sample index:

```
t_hat_n = t_anchor + (n - n_anchor) / fs_eff
fs_eff = (n_2 - n_1) / (t_2 - t_1)
```

where (n_1, t_1) and (n_2, t_2) are sample-count / LSL-time anchor pairs. The reconstructed axis is used only when it improves consistency and is logged as a timing caveat.

11.3 Branch-relative time

For rendering and anchor analysis:

```
t_rel = t_abs - t_phase_start
```

Event overlays should be repaired when mixed timebases are detected:

```
if event_time > phase_duration and phase_start_abs is known:
event_time_rel = event_time - phase_start_abs
else:
event_time_rel = event_time
```

12. Core EEG spectral features

SHARED FEATURE LAYER

12.1 Filtering

For a band [f1, f2], a band-limited signal is:

```
x_band,ch(t) = BandPass(x_ch(t), f1, f2)
```

12.2 Welch power spectral density

For window W_t:

```
PSD_ch(f,t) = WelchPSD(x_ch over W_t)
BandPower_ch,b(t) = integral_{f in band b} PSD_ch(f,t) df
LogBandPower_ch,b(t) = log(BandPower_ch,b(t) + eps)
```

12.3 ROI aggregation

For ROI R with channels ch in R:

```
BandPower_R,b(t) = mean_{ch in R}(LogBandPower_ch,b(t))
```

or robust median:

```
BandPower_R,b(t) = median_{ch in R}(LogBandPower_ch,b(t))
```

12.4 Phase and phase-locking value

Using the analytic signal:

```
z_ch,b(t) = Hilbert(x_band,ch(t))
phase_ch,b(t) = angle(z_ch,b(t))
```

Pairwise phase-locking:

```
PLV_{a,b}(W) = | (1/N) * sum_{t in W} exp(i*(phase_a(t) - phase_b(t))) |
```

ROI PLV:

```
PLV_ROI(W) = mean_{channel pairs in ROI}(PLV_pair(W))
```

13. ArtifactScore

VETO AND COVARIATE LAYER

ArtifactScore is a veto/covariate layer, not a biological finding.

13.1 Peak-to-peak amplitude

```
p2p_ch(W) = max_{t in W}(x_ch(t)) - min_{t in W}(x_ch(t))
```

13.2 High-frequency sentinel

A generic muscle/line-noise sentinel:

```
HF_ch(W) = log( integral_{45 Hz}^{55 Hz} PSD_ch(f,W) df + eps )
```

Optional wider EMG sentinel:

```
EMG_ch(W) = log( integral_{55 Hz}^{95 Hz} PSD_ch(f,W) df + eps )
```

13.3 Ocular/frontal sentinel

For frontal channel group F:

```
FpSentinel(W) = mean_{ch in F}( z(p2p_ch(W)) + z(HF_ch(W)) ) / 2
```

13.4 Composite artifact score

Default operational form:

```
ArtifactScore(W) = mean(
rz(p2p_all(W)),
rz(HF_45_55(W)),
rz(EMG_55_95(W)),
rz(FpSentinel(W)),
rz(line_noise_ratio(W)),
rz(flatline_or_step_flag(W))
)
```

A stricter gate uses:

```
ArtifactPass(W) = 1[ArtifactScore(W) < theta_artifact]
```

For peak modules:

```
QC_j = ArtifactPass(W_hit,j) * TimingPass_j * ConfoundPass_j
```

14. ALS timing, preflight, and script-location test

REQUIRED TIMING VERIFICATION

```
ALS_pulse(t) = 1 during the declared display pulse; 0 otherwise
```

The separate 30-second preflight reports effective sample rate, timestamp monotonicity, gaps, flat channels, saturation/rail contact, noise scale, and ALS pulse visibility. It is a user-invoked validation action, not an automatic block on arming unless a study configuration explicitly makes it mandatory.

The barcode placement script-location test verifies the actual launch path, expected script, runner registration, visible pulse, and recovered timing relationship. It validates timing infrastructure, not neural theory.

> ALS boundary
> The photodiode/ALS signal belongs to the exogenous stimulus vector u(t); it is never biological hidden-state evidence.

15. Continuous Autonomic/RespDualPath v1.0

IMPLEMENTED EXPLORATORY MODULE

> Primary rule
> Interpolated HR supports display and alignment. HRV is computed only from retained beat events. Autonomic outputs remain separate from primary CAI until independently validated.

15.1 Cardiac beat events and grid

```
IBI_j = RR_j / 1000 seconds
HR_j = 60 / IBI_j = 60000 / RR_j bpm
300 ms <= RR_j <= 2000 ms
```

Valid instantaneous HR is interpolated to 4 Hz for alignment, with interpolation flags. Maximum nearest-beat distance is 3 seconds and the surrounding gap is at most 6 seconds.

15.2 Beat-derived rolling HRV

```
RMSSD_W = sqrt[(1/Ndiff) sum_j (RR_(j+1)-RR_j)^2]
SDNN_W = sqrt[(1/(N-1)) sum_j (RR_j-mean(RR))^2]
pNN50_W = (1/(N-1)) sum_j 1(|RR_(j+1)-RR_j| > 50 ms)
```

Centered 30-second RMSSD requires at least 15 valid differences. Centered 60-second SDNN/pNN50 requires at least 30 valid beats. Coverage and exclusion fractions are reported.

15.3 Respiration

```
r_f(t) = zero-phase Butterworth bandpass_[.05,.70 Hz](r_raw(t))
z_r(t) = r_f(t) + i H[r_f(t)]
phase_r(t) = unwrap(angle(z_r(t)))
Depth(t) = 2|z_r(t)|
BR(t) = (60/(2*pi)) d phase_r(t)/dt
velocity(t) = d r_f(t)/dt
acceleration(t) = d^2 r_f(t)/dt^2
```

Breathing rate is smoothed with a centered five-second median and retained within 2-60 breaths/minute. A sigh candidate is a robust amplitude peak >= 3.0 with an eight-second refractory interval. Motion caution is max(|z velocity|,|z acceleration|) > 4.0.

Implementation audit

The source theory described hold as low amplitude AND low velocity for at least four seconds. v1.0 currently uses low amplitude OR near-zero velocity. Hold labels are permissive candidates and require sensitivity analysis.

15.4 HR-respiration coupling

```
r(lag;t) = corr_Wt(HR(u), respiration_filtered(u+lag))
lag in {-10.00,-9.75,...,9.75,10.00} seconds
lag_star(t) = argmax_lag |r(lag;t)|
p_shift = [1 + sum_b 1(|r_b_star| >= |r_obs_star|)]/(B+1)
B=200 circular shifts; minimum shift=30 seconds
```

The best-lag search is exploratory and multiplicity-sensitive. Coherence, HR-respiration PLV, regulatory-recovery composites, and R_auto remain future formulas and do not enter primary CAI.

16. API_A v1: autonomic availability index

SECONDARY PHYSIOLOGICAL SUMMARY

API_A is a proxy for regulated availability, not valence, clinical calm, or proof of parasympathetic healing.

16.1 HR and HRV features

Given R-R intervals RR_k in milliseconds:

```
HR_k = 60000 / RR_k
RMSSD = sqrt( mean( (RR_{k+1} - RR_k)^2 ) )
SDNN = std(RR_k)
pNN50 = count(|RR_{k+1} - RR_k| > 50 ms) / count(pairs)
```

HR slope over a window:

```
HR_slope(W) = slope of linear regression HR(t) ~ t over W
```

16.2 API_A formula

Default operational form:

```
API_A(t) = 0.30 * z(-HR_mean(t))
+ 0.25 * z(RMSSD(t))
+ 0.20 * z(SDNN(t))
+ 0.15 * z(pNN50(t))
+ 0.10 * z(-HR_slope(t))
- 0.10 * z(RespInstability(t))
```

If some HRV features are unavailable, weights should be renormalized over available features.

Interpretation:

```
higher API_A = lower HR / higher HRV / more regulated physiology, after respiration
caution
```

PART IV
Primary recognition and integration path

Separates candidate recognition timing from delayed integration and applies locked-anchor quality gates.

17. GammaScalpel

ARTIFACT-SENSITIVE WORK-SIGNAL FAMILY

GammaScalpel treats lower-gamma as an artifact-sensitive work-signal family, not a meaning biomarker.

17.1 Microband decomposition

Define 5 Hz bands:

```
B_1 = 30-35 Hz
B_2 = 35-40 Hz
B_3 = 40-45 Hz
B_4 = 45-50 Hz  (more artifact-sensitive)
B_5 = 50-55 Hz  (line / sentinel caution)
```

For each ROI R and band b:

```
Gamma_R,b(t) = rz(LogBandPower_R,b(t))
```

17.2 MeaningGamma proxy

MeaningGamma is a weighted candidate signal. It should not be interpreted without artifact and stimulus controls.

```
MeaningGamma(t) = sum_b w_b * [
a_PT  * Gamma_posterior_temporal,b(t)
+ a_P   * Gamma_parietal,b(t)
+ a_O   * Gamma_occipital,b(t)
- a_F   * Gamma_frontal_task,b(t)
]
- lambda_art * ArtifactScore(t)
- lambda_vis * VisualDrive(t)
- lambda_cue * CueOn(t)
```

Typical interpretation:

```
posterior/temporal + parietal lower-gamma -> candidate semantic/scene recognition
work
frontal/task gamma -> analytic extraction or task stance
visual/cue/artifact terms -> non-meaning penalties
```

17.3 TaskGamma proxy

```
TaskGamma(t) = sum_b w_b * [
a_F  * Gamma_frontal,b(t)
+ a_C  * Gamma_central,b(t)
+ a_PLV * PLV_fronto_central,b(t)
]
- lambda_art * ArtifactScore(t)
```

TaskGamma is expected to rise during Contextual Override when the subject is doing the running-sum task.

17.4 GammaScalpel band decision

A band is considered interpretable only if:

```
BandPass_b = 1[Artifact_b < threshold]
* 1[LineNoise_b < threshold]
* 1[VisualDrive_b not sufficient explanation]
```

Then:

```
MeaningGamma_valid(t) = MeaningGamma(t) * BandPass(t)
```

18. Temporal Semantic Proxy (TSP)

CANDIDATE TIMING PROXY

TSP estimates a candidate time of semantic recognition. It is not a ground-truth reading of meaning.

18.1 Components

```
Gm(t) = MeaningGamma(t)
PT(t) = posterior-temporal semantic-work proxy
N(t)  = NAST / absorption-transition support, if available
K(t)  = local KHT-topo support, if available
A(t)  = ArtifactScore(t)
V(t)  = VisualDrive(t)
X(t)  = TaskGamma(t)
```

18.2 TSP formula

```
TSP(t) = w_G  * rz(Gm(t))
+ w_PT * rz(PT(t))
+ w_N  * rz(N(t))
+ w_K  * rz(K(t))
- w_X  * rz(X(t))
- w_A  * rz(A(t))
- w_V  * rz(V(t))
```

In many runs, TSP is treated as:

```
TSP_z(t) = rz(TSP(t))
```

18.3 Event peak

For anchor window W_j:

```
t_peak,j = argmax_{t in W_j}(TSP_z(t))
TSP_peak,j = max_{t in W_j}(TSP_z(t))
```

A strict event should not rely only on TSP. TSP identifies candidate timing; MRED/A-MRED tests whether recognition and integration survive the gates.

19. Theta handoff / integration proxy

CANDIDATE DELAYED-INTEGRATION FEATURE

19.1 Theta feature

Theta_R(t) = rz(log integral_{4 Hz}^{8 Hz} PSD_R(f,t) df) Use frontal-midline / frontocentral / parietal integration ROI as configured.

19.2 Post-event theta delta

For event time t_j:

Theta_pre,j = mean(Theta_R(t) over [t_j - 10, t_j]) Theta_post_0_10,j = mean(Theta_R(t) over [t_j, t_j + 10]) Theta_post_10_30,j = mean(Theta_R(t) over [t_j + 10, t_j + 30])

```
H_theta_0_10,j = Theta_post_0_10,j - Theta_pre,j
```

```
H_theta_10_30,j = Theta_post_10_30,j - Theta_pre,j
```

19.3 Handoff criterion

ThetaHandoff_j = 1[H_theta_10_30,j > theta_handoff_threshold] The delayed 10-30 s window is often more relevant than the immediate 0-10 s window when testing reflective integration rather than sensory surprise.

20. CandidateLocal KHT-topo

EXPLORATORY RECIPROCAL MODEL

CandidateLocal_KHT-topo is an event-level coupling model.

20.1 Human-scale state variables

```
E_t = observed work-signal proxy, e.g. MeaningGamma or TSP
Y_t = inferred human-translation state proxy, e.g. theta/topology/NIP state
u_t = exogenous stimulus regressors
A_t = artifact/confound regressors
```

20.2 Local reciprocal model

For local event window W_j:

```
E_{t+Delta} = a_E * E_t + eta_HT * Y_t + b_E^T u_t + c_E * Artifact_t + eps_E,t
Y_{t+Delta} = a_Y * Y_t + zeta_HT * E_t + b_Y^T u_t + c_Y * Artifact_t + eps_Y,t
where:
eta_HT  = Y -> E coefficient
zeta_HT = E -> Y coefficient
```

20.3 Local coupling coefficient

```
K_HT_topo,j = sqrt( abs(eta_HT,j * zeta_HT,j) )
LoopSign_j = sign(eta_HT,j * zeta_HT,j)
```

20.4 Event-lock rule

```
KLock_j = 1[K_HT_topo,j >= threshold_K or percentile_K >= threshold_percentile]
EventLock_j = KLock_j * ThetaHandoff_j * ArtifactPass_j * TimingPass_j
```

K alone is not enough. Theta handoff alone is not enough.

20.5 Model-selection guard

For candidate models:

```
M_full = reciprocal E <-> Y
M_replay = replay-only / no reciprocal hidden loop
M_eta = eta-only
M_zeta = zeta-only
M_null = no-coupling
M_generic = generic hidden oscillator
```

Use held-out prediction:

```
CVWin_j = 1[CVError_full < min(CVError_alternatives)]
```

A strong candidate requires:

```
EventLock_j * CVWin_j = 1
```

21. MRED: Meaning Recognition / Encoding Dissociation

PRIMARY CONSTRUCT
DECOMPOSITION

MRED separates meaning recognition from integration/carryover.

21.1 Meaning recognition score

```
MR_j = w_TSP * TSP_peak,j
+ w_G   * MeaningGamma_peak,j
+ w_K   * KHT_topo_j
+ w_N   * NAST_j
- w_A   * ArtifactScore_j
- w_V   * VisualDrive_j
- w_C   * ConfoundBurden_j
```

21.2 Encoding / integration score

```
ENC_j = w_H  * H_theta_10_30,j
+ w_T  * TopoShift_j
+ w_E  * EET_echo_j
+ w_API * API_A_post,j
+ w_B2 * Baseline2Echo_j
- w_A  * ArtifactScore_post,j
- w_X  * TaskGamma_post,j
```

21.3 MRED quadrant

Given thresholds theta_MR and theta_ENC:

```
if MR_j >= theta_MR and ENC_j >= theta_ENC:
quadrant = MR_HIGH_ENC_HIGH
elif MR_j >= theta_MR and ENC_j < theta_ENC:
quadrant = MR_HIGH_ENC_LOW
elif MR_j < theta_MR and ENC_j >= theta_ENC:
quadrant = MR_LOW_ENC_HIGH
else:
quadrant = MR_LOW_ENC_LOW
```

Interpretation:

```
MR_HIGH_ENC_HIGH = recognition plus integration candidate
MR_HIGH_ENC_LOW  = recognition without clear integration; can occur with familiarity
MR_LOW_ENC_HIGH  = possible nonsemantic update, task/cue/respiration/artifact
caution
MR_LOW_ENC_LOW   = no detected event under current criteria
```

22. A-MRED: anchor-locked endpoint

PRIMARY CANDIDATE GATE

A-MRED is the compressed primary endpoint.

22.1 Strict gate

For a predeclared anchor j:

```
A_MRED_j =
1[AnchorLocked_j = 1]
* 1[MR_T,j  > theta_MR]
* 1[ENC_T,j > theta_ENC]
* 1[MR_T,j  > MR_C,j + delta_MR]
* 1[MR_T,j  > MR_O,j + delta_MR]
* 1[ENC_T,j > ENC_C,j + delta_ENC]
* 1[ENC_T,j > ENC_O,j + delta_ENC]
* 1[QC_j = PASS]
where:
QC_j = ArtifactPass_j * TimingPass_j * ConfoundPass_j * CETRPass_j
```

22.2 Continuous backup score

```
MREDScore_condition,j = clip_plus(MR_condition,j) * clip_plus(ENC_condition,j) *
QC_condition,j
DeltaMRED_j = MREDScore_T,j - max(MREDScore_C,j, MREDScore_O,j)
```

22.3 Claim-grade modifiers

STRICT_CONFIRMATORY:

anchor locked before acquisition, frame verified, runner registered, ALS/timing pass.

RUNNER_REGISTERED_ESTIMATED:

anchor loaded before acquisition but estimated/not frame verified.

CONCEPTUALLY_PREDECLARED:

anchor existed in planning but not machine registered.

EXPLORATORY:

event found after physiology review.

23. MRED-Peak and MRED-Resolution

PRIMARY CANDIDATE PROFILES

This module separates acute meaning shock from delayed reflective closure.

23.1 MRED-Peak

MRED-Peak asks whether an anchor produces acute Target-specific recognition and delayed integration.

```
PeakEvidence_j = mean(
z(MR_T,j),
z(ENC_T,j),
z(DeltaMRED_j),
z(CII_margin_j),
z(KHT_topo_j)
)
```

Strict peak pass:

```
MRED_Peak_Pass_j = A_MRED_j or BIT_j
```

Soft peak candidate:

```
MRED_Peak_Candidate_j = 1[PeakEvidence_j > theta_peak] * 1[QC_j != FAIL]
```

23.2 MRED-Resolution

MRED-Resolution asks whether the event produces delayed reflective/regulatory recovery rather than a sharp event spike.

```
ResolutionEvidence_j = mean(
z(Baseline2Echo_j),
z(EET_j),
z(RDI_j),
z(TargetEchoedDuringBaseline2_report),
z(EmotionalAfterglow_T - EmotionalAfterglow_O),
z(API_A_B2 - API_A_B1)
)
```

Resolution candidate:

```
MRED_Resolution_Candidate_j = 1[ResolutionEvidence_j > theta_resolution]
* 1[TargetSpecificity_j = 1]
* 1[QC_j != FAIL]
```

23.3 Run-level archetype

```
if max(PeakEvidence) high and ResolutionEvidence moderate:
archetype = MRED_PEAK
elif ResolutionEvidence high and PeakEvidence moderate:
archetype = MRED_RESOLUTION
elif MR high and ENC low:
archetype = RECOGNITION_DOMINANT
else:
archetype = UNRESOLVED
```

PART V
Secondary summaries and decoder context

Summarizes continuous immersion, task tradeoffs, update polarity, decoder availability, and report correspondence.

24. NIP: Narrative Immersion Proxy

SECONDARY SUMMARY

NIP operationalizes attention plus resonance without claiming dopamine, oxytocin, or proprietary immersion biology.

24.1 Semantic attention component

```
A_sem(t) = clip_plus(
w_G * rz(MeaningGamma(t))
+ w_T * rz(TSP(t))
+ w_N * rz(NAST(t))
- w_V * rz(VisualDrive(t))
- w_A * rz(ArtifactScore(t))
- w_X * rz(TaskGamma(t))
)
```

24.2 Integration / resonance component

```
R_int(t) = clip_plus(
w_H   * rz(H_theta(t))
+ w_K   * rz(KHT_topo(t))
+ w_TOPO* rz(TopoShift(t))
+ w_API * rz(API_A(t))
- w_A   * rz(ArtifactScore(t))
)
```

24.3 NIP density

With lag ell:

```
NIP_density(t) = A_sem(t)^alpha * R_int(t + ell)^beta * Penalty(t)
where:
Penalty(t) = ArtifactPass(t) * ConfoundPenalty(t) * TimingPass(t)
```

Default:

```
alpha = 1
beta = 1
```

25. BIT: Bivariate Immersion Threshold

SECONDARY EVENT RULE

BIT is an AND-gate, not an average.

```
BIT_j = 1[A_sem_j > theta_A]
* 1[R_int_j > theta_R]
* 1[Artifact_j < theta_artifact]
* 1[Specificity_j = 1]
where:
```

```
Specificity_j = 1[Target_j > Control_j + delta]
* 1[Target_j > Override_j + delta]
```

BIT fails when recognition is high but integration is absent:

```
MR_HIGH_ENC_LOW -> BIT = 0
```

This prevents a high attention spike from being mislabeled as full immersion.

26. CII: Continuous Immersion Index

SECONDARY ANCHOR INTEGRAL

CII measures average immersion density across an anchor window.

```
CII_condition(W_j) = (1 / |W_j|) * integral_{t in W_j} NIP_density_condition(t) dt
```

Discrete implementation:

```
CII_condition(W_j) = mean_{t in W_j}(NIP_density_condition(t))
```

Target-specific margin:

```
CII_margin_j = CII_T,j - max(CII_C,j, CII_O,j)
```

Duration normalization matters:

without division by |W_j|, long scenes automatically score higher.

27. IAQ / ITQ: Immersion Attenuation Quotient

SECONDARY TARGET-OVERRIDE SUMMARY

IAQ estimates how much Contextual Override attenuated Target immersion.

```
IAQ_j = 1 - (CII_O,j + eps) / (CII_T,j + eps)
```

Equivalent difference form:

```
ITQ_j = CII_T,j - CII_O,j
```

Interpretation:

```
IAQ near 1  = Override strongly attenuated the immersion proxy
IAQ near 0  = Override preserved a similar immersion proxy
IAQ below 0 = Override exceeded Target on the immersion proxy
```

Use IAQ in technical documents; reserve "theft" language for clearly labeled philosophical/cultural discussion.

28. TTI: Reception-Extraction Tradeoff

SECONDARY TASK COMPARISON

TTI estimates whether the same narrative was received or converted into a task object.

28.1 Budget principle

```
c_R R(t) + c_X X(t) + c_M M(t) + c_Gamma Gamma(t) + c_RX R(t)X(t) <= B(t)
where:
R(t) = receptive absorption
X(t) = extractive / instrumental task stance
M(t) = semantic payload
Gamma(t) = cognitive/metabolic exhaust proxy
B(t) = finite processing budget
c_RX R(t)X(t) = collision cost
```

28.2 Dynamic model

```
dR/dt = alpha_M*M(t)*(1-R) + alpha_A*A(t)
- beta_X*X(t)*R(t) - beta_Gamma*Gamma(t)*R(t) - lambda_R*R(t)
dX/dt = alpha_T*T(t)*(1-X) + alpha_C*Cue(t)
- beta_R*R(t)*X(t) - lambda_X*X(t)
```

28.3 TTI composite

```
TTI = w1 * z(MR_T - MR_O)
+ w2 * z(ENC_T - ENC_O)
+ w3 * z(API_T - API_O)
+ w4 * z(X_O - X_T)
- w5 * z(|Artifact_T - Artifact_O|)
- w6 * z(|VisualDrive_T - VisualDrive_O|)
```

Default weights:

```
w1 MR       = 0.28
w2 ENC      = 0.24
w3 API_A    = 0.14
w4 X-load   = 0.22
w5 Artifact = 0.08 penalty
w6 Visual   = 0.04 penalty
```

Positive TTI means Target preserved more reception/integration while Override carried more extraction/task load.

29. NUPI: Narrative Update Polarity Index

SECONDARY UPDATE PROFILE

NUPI asks what kind of update meaning demanded.

29.1 Accommodative Load Index (ALI)

```
ALI = mean(
z(semantic_intensity),
z(Target_specificity),
z(complexity_perturbation),
z(TTI_positive_load),
z(primary_endpoint_pass_strength)
)
```

Interpretation:

```
High ALI = model-expansion, awe, shock, destabilizing accommodation, or high update
```

demand.

29.2 Regulatory Dividend Index (RDI)

```
RDI = mean(
z(Baseline2_regulation),
z(Baseline2_semantic_echo),
z(EET_afterstate_echo),
z(TargetEchoedDuringBaseline2_selfreport),
z(API_A_B2 - API_A_B1)
)
```

Interpretation:

```
High RDI = reflective recovery, closure, parasympathetic regulation, after-state
```

coherence.

29.3 NUPI score

```
NUPI = RDI - ALI
```

Classification:

```
NUPI > theta_positive and RDI high:
RESOLUTIVE_RECOVERY
```

ALI high and RDI moderate/high:

HIGH_LOAD_WITH_RECOVERY ALI high and RDI low:

ACCOMMODATIVE_LOAD MR high but ENC/RDI unclear:

RECOGNITION_DOMINANT_OR_UNRESOLVED Boundary:

NUPI does not measure heat, ATP, clinical healing, morality, or literal thermodynamics.

30. DGA: Decoder Gate Availability

SECONDARY ATTRIBUTION MODEL

30.1 Rationale

DGA formalizes a dissociation exposed by the Eternal Sunshine run: semantic clarity and emotional integration can separate across branches. A Target can be emotionally absorbing despite degraded audio access; an Override can provide clearer semantic access while still imposing analytic extraction load.

The model rejects the simple stimulus-transfer equation:

```
MeaningResponse = f(stimulus)
```

and replaces it with:

```
MeaningResponse = f(stimulus_geometry, semantic_access, decoder_availability,
emotional_integration, extraction_load, confound_burden,
prior_state_space)
```

30.2 Branch-level components

For branch b, define semantic access:

```
SA_b = 1 - mean(AudioComprehensionDifficulty_b,
SpeakerVolumeDifficulty_b,
ExternalNoiseIntrusion_b,
AudioVideoSyncProblem_b) / 9
```

Decoder availability:

```
DA_b = mean(Meaning_b,
Absorption_b,
EmotionalAfterglow_b,
StoryActiveWashout_b) / 9
```

Emotional/integrative impact:

```
EI_b = mean(Absorption_b,
EmotionalAfterglow_b,
StoryActiveWashout_b) / 9
```

Extraction load:

```
X_b = max(TaskExtractionLoad_core_b,
OverrideTaskBurden_b) / 9
```

Confound burden:

```
C_b = max(ConfoundBurden_core_b,
DetailedConfoundMean_b) / 9
```

30.3 Gate equation

The branch-level decoder-gate estimate is:

```
D_gate,b = sigmoid(2.0*SA_b + 1.4*DA_b + 1.1*EI_b
- 1.2*X_b - 1.8*C_b - 0.75)
where:
sigmoid(x) = 1 / (1 + exp(-x))
```

High D_gate means the branch had good semantic access, strong subjective availability/impact, and tolerable task/confound burden. Low D_gate means one or more gates were closed.

30.4 Gate Dissociation Index

The Gate Dissociation Index compares Target and Override:

```
GDI = mean(SA_Override - SA_Target,
EI_Target - EI_Override,
X_Override - X_Target,
C_Target - C_Override)
```

A high positive GDI means:

Override had better semantic access, Target had stronger emotional/integrative impact, Override had greater extraction load, Target had greater confound burden.

This is the signature of a gate-dissociation run.

30.5 DGA labels

TARGET_GATE_OPEN_PEAK_PROFILE:

Target gate is high, Target specificity is high, and MRED-Peak is supported.

TARGET_GATE_OPEN_RESOLUTION_PROFILE:

Target gate is high, resolution/afterglow/Baseline2 evidence is strong.

DECODER_GATE_DISSOCIATION_AUDIO_ACCESS_SPLIT:

semantic access and emotional impact split across Target and Override.

FEATURE_PROXY_RECOGNITION_DOMINANT_GATE_UNRESOLVED:

older run lacks PRAYCG2.0 reports/Baseline2; only feature-level interpretation is possible.

NO_CLEAR_GATE_OPENING:

neither Target nor Override has sufficient availability or specificity.

30.6 DGA is secondary

DGA does not replace A-MRED. DGA explains why A-MRED may pass, fail, or split. It belongs in the secondary tier:

PRIMARY:

```
Timing/QC -> CET-R -> A-MRED / MRED-Peak-Resolution
```

SECONDARY:

NIP / TTI / NUPI / DGA EXPLORATORY:

KHT-topo / NAST / EET / MRED-ITP / OCM / RSM / CVB / Squint

31. Persistent learned state-space formulation

HUMAN-SCALE THEORY MODEL

31.1 Persistent learned state-space

The decoder-gate update implies that a person does not enter the chamber as a blank slate. The viewer arrives with a learned state-space:

```
P_state(t) = [autobiographical traces, semantic priors, attachment models,
```

current-life relevance, prior exposure, affective schemas] Incoming stimulus geometry is encoded as:

phi(u(t)) Meaning recognition is then:

```
MR(t) = D_gate(t) * sim(phi(u(t)), P_state(t))
```

Integration requires later availability:

```
ENC(t + tau) = MR(t) * A_reg(t + tau) * [1 - X_extract(t + tau)] * Q_carry(t + tau)
```

31.2 Update equation

If an event is integrated, the state-space may update:

```
P_state(t+1) = P_state(t) + eta_update * MR(t) * ENC(t+tau) * Relevance(t) - decay/noise
```

This is a human-scale topology equation, not a cellular proof.

31.3 Consequence for stimulus interpretation

A low-bandwidth cue can have large physiological consequence if it fits a high-relevance internal attractor:

LargeResponse possible when sim(phi(u), P_state) is high even if sensory bandwidth is low This explains why partial semantic access may still open a strong decoder gate in a familiar or personally resonant stimulus.

32. Baseline 1 versus Baseline 2

SECONDARY CUMULATIVE COMPARISON

32.1 Feature delta

For feature F:

```
DeltaF_B2_B1 = mean(F over B2) - mean(F over B1)
```

32.2 Reflective-state index

```
ReflectiveIndex = mean(
z(DeltaTSP_B2_B1),
z(DeltaNIP_B2_B1),
z(DeltaAPI_A_B2_B1),
z(-DeltaTaskGamma_B2_B1),
z(-DeltaArtifact_B2_B1)
)
```

32.3 Regulation index

```
Baseline2Regulation = mean(
z(-DeltaHR_B2_B1),
z(DeltaRMSSD_B2_B1),
z(DeltaSDNN_B2_B1),
z(DeltaPNN50_B2_B1),
z(-DeltaRespInstability_B2_B1)
)
```

These feed MRED-Resolution and NUPI.

33. SelfReport20 parser

INDEPENDENT EVIDENCE STREAM

Self-report is an independent evidence stream, not proof.

33.1 Branch core report vector

```
Report_branch = [Meaning, Absorption, EmotionalAfterglow, StoryActiveWashout,
TaskExtractionLoad, ConfoundBurden]
```

33.2 Final report vector

```
Report_final = [OverrideReducedReception, StoryBrokeThroughOverride,
```

TargetEchoedDuringBaseline2, Familiarity, NewMeaningToday, CurrentLifeResonance]

33.3 Model relation

A generic mixed model:

```
Report_{i,k} = alpha
```

```
+ beta_MR * MR_{i,k}
+ beta_ENC * ENC_{i,k}
+ beta_NIP * NIP_{i,k}
+ beta_API * API_A_{i,k}
- beta_X * TaskGamma_{i,k}
- beta_A * Artifact_{i,k}
+ u_subject
+ eps_{i,k}
```

A positive correspondence is not proof of internal state; contradiction is a warning.

PART VI
Exploratory and convergence modules

Provides candidate mechanisms, entrainment controls, and development indices that remain isolated from primary eligibility.

34. NAST: Narrative Absorption State Transition

EXPLORATORY TRANSITION MODEL

NAST estimates transition from idle/analytic/resting state into narrative absorption.

34.1 Alpha/idling proxy

```
Alpha_R(t) = rz(log integral_{8 Hz}^{12 Hz} PSD_R(f,t) df)
```

Absorption is often expected to show alpha desynchronization relative to baseline, but this is not universal.

34.2 NAST proxy

```
NAST(t) = w_TSP * rz(TSP(t))
+ w_G   * rz(MeaningGamma(t))
+ w_H   * rz(ThetaIntegration(t))
- w_A   * rz(Alpha(t))
- w_X   * rz(TaskGamma(t))
- w_art * rz(ArtifactScore(t))
```

34.3 Phase transition contrast

For transition from phase p to phase q:

```
DeltaNAST_{p->q} = mean(NAST over early q) - mean(NAST over late p)
```

Example:

```
WASHOUT_1 -> TARGET
WASHOUT_2 -> OVERRIDE
```

Interpretation:

```
positive Target transition = narrative absorption candidate
positive Override transition with TaskGamma = analytic reorientation candidate
```

35. OCM025: Override Cue Microstate analysis

EXPLORATORY CUE ANALYSIS

OCM025 analyzes cue-locked micro-windows at 0.25 s resolution.

35.1 Cue-relative bins

For cue onset t_i:

```
pre_i       = [t_i - 0.50, t_i)
digit_i     = [t_i, t_i + 0.85]
update_i    = [t_i + 0.25, t_i + 1.25]
maint_i     = [t_i + 1.25, min(t_i + 3.0, t_{i+1}))
```

35.2 Digit recognition gamma

```
DR_i = mean(MeaningGamma or visual/digit gamma over digit_i)
- mean(same feature over pre_i)
```

35.3 Working-memory update theta

```
WMU_i = mean(frontal_theta over update_i) - mean(frontal_theta over pre_i)
```

35.4 Maintenance theta

```
MAINT_i = mean(frontal_theta over maint_i) - mean(frontal_theta over pre_i)
```

35.5 OCM score

```
OCM_i = 0.35*z(DR_i)
+ 0.35*z(WMU_i)
+ 0.20*z(MAINT_i)
- 0.10*z(Artifact_i)
```

OCM is Override-only unless explicitly used as a cue confound check.

36. RSM: Running Sum Microstate Model

EXPLORATORY TASK-STATE MODEL

RSM estimates arithmetic/cue burden during Override.

36.1 Running sum

```
S_i = S_{i-1} + v_i
```

36.2 Carry features

```
prior_units_i = S_{i-1} mod 10
carry_required_i = 1[prior_units_i + v_i >= 10]
high_carry_load_i = max(0, prior_units_i + v_i - 9)
hard9_i = 1[v_i = 9 and prior_units_i in {7,8,9}]
```

36.3 Arithmetic Compute Load

```
ACL_i = beta1*z(log(1 + S_{i-1}))
+ beta2*carry_required_i
+ beta3*z(high_carry_load_i)
+ beta4*hard9_i
+ beta5*z(fatigue_or_elapsed_time_i)
```

36.4 Compute-stall score

```
ComputeStall_i = 0.35*z(WMU_i)
+ 0.30*z(MAINT_i)
+ 0.25*z(T_close_i)
+ 0.10*OpenLoop_i
- 0.20*z_positive(Artifact_i)
where:
T_close_i = time until theta/update loop returns below closure threshold
OpenLoop_i = 1[T_close_i > t_{i+1} - safety_buffer]
```

36.5 Latent guess-risk

```
GuessRisk_i = sigmoid(
gamma0
+ gamma1*ArithmeticStall_i
+ gamma2*DeadlinePressure_i
+ gamma3*ACL_i
+ gamma4*CueVisibilityBurden_i
+ gamma5*PriorUncertainty_i
- gamma6*FinalSumConfidence
)
```

This is a latent proxy, not proof that the subject guessed.

37. CVB: Cue Visibility / Legibility Burden

CONFOUND DIAGNOSTIC

CVB estimates cue-reading strain.

```
CVB_i = 0.45*z(HoldBurden_i)
- 0.25*z(DigitRecognition_i)
+ 0.15*z(VisualSearch_i)
+ 0.15*z(1 - CueContrast_i)
+ 0.15*z(CueBlurOrSmallness_report)
- 0.10*z(Artifact_i)
```

For small/blurred cues, contrast may be high while legibility remains poor. CVB should therefore not rely only on luminance contrast.

38. SquintProxy

CONFOUND DIAGNOSTIC

SquintProxy estimates frontal visual-strain artifacts. It cannot confirm squinting.

```
SquintProxy_i = 0.30*z(Fp1_HF_i)
+ 0.25*z(Fp1_p2p_i)
+ 0.20*z(frontal_EMG_proxy_i)
+ 0.15*z(EyeStrainOrSquint_report_i)
+ 0.10*z(CueBlurOrSmallness_report_i)
- 0.20*z(BlinkTemplate_i)
```

Interpretation:

```
high SquintProxy = possible visual strain / ocular or forehead artifact
not confirmed squint
```

EOG or eye tracking is required for confirmation.

39. Confound registry

REQUIRED WARNING REGISTRY

Confound reports are covariates/veto aids.

39.1 Branch confound score

```
ConfoundBurden_branch = mean(
AudioVideoSyncProblem,
AudioComprehensionDifficulty,
ExternalNoiseIntrusion,
SpeakerVolumeDifficulty,
CueBlurOrSmallness,
EyeStrainOrSquint
) / 9
```

39.2 Gate

```
ConfoundPass_branch = 1[ConfoundBurden_branch < theta_confound]
```

or claim-grade modifier:

```
if ConfoundBurden_branch >= theta_caution:
claim_level = claim_level + "_CONFOUND_CAUTION"
```

Confounds should lower claim strength, not create positive evidence.

40. CET: Cinematic Entrainment Tracking

STIMULUS-TRACKING DIAGNOSTIC

CET quantifies how much EEG tracks exogenous stimulus rhythm.

40.1 Stimulus transform

```
U_m(f,tau) = STFT[u_m(t)]
```

where u_m may be audio envelope, luminance, visual change, cut train, cue train, etc.

40.2 EEG transform

```
X_ch(f,tau) = STFT[x_ch(t)]
```

40.3 Coherence

```
Coh_{u,x}(f,tau) = |S_{u,x}(f,tau)|^2 / (S_{u,u}(f,tau) * S_{x,x}(f,tau) + eps)
```

40.4 Phase lag

```
phi_{u,x}(f,tau) = angle(S_{u,x}(f,tau))
lag_{u,x}(f,tau) = phi_{u,x}(f,tau) / (2*pi*f)
```

Caution:

Stable lag can mean good tracking, not stalled biological momentum.

41. CET-R: residualization against exogenous drive

ATTRIBUTION GATE

CET-R asks whether MRED/NIP/TTI survives after modeling stimulus rhythm.

41.1 Regression model

For feature Y(t) such as NIP density, MeaningGamma, TSP, MR, or ENC:

```
Y(t) = beta0 + beta_u^T U(t) + beta_art*Artifact(t) + eps(t)
where:
U(t) = [audio_env, luminance, visual_change, optical_flow, cut_train,
```

flash_train, cue_on, cue_value, cue_phase_sin, cue_phase_cos, ALS_pulse] Residual:

```
Y_resid(t) = Y(t) - Y_hat(t)
```

41.2 Blocked cross-validation

Split time into blocks, fit on training blocks, evaluate held-out blocks:

```
R2_blocked = 1 - sum((Y_test - Y_hat_test)^2) / sum((Y_test - mean(Y_train))^2)
```

If blocked-CV is poor or negative, the stimulus regressors do not generalize as an explanation.

41.3 Residualized CII

```
CII_resid(W_j) = mean_{t in W_j}(clip_plus(Y_resid(t)))
```

A robust Target effect should remain Target-dominant after residualization.

42. EET: Endogenous Echo Tracking

EXPLORATORY PERSISTENCE MODEL

EET asks whether a stimulus-anchor state vector resembles later washout/Baseline2 physiology.

42.1 State vector

For anchor j:

```
StateVec_anchor,j = [MR, ENC, NIP, TSP, MeaningGamma, Theta, Alpha, API_A, HRV, Resp,
...]_Wj
```

For after-state window q:

```
StateVec_after,q = [same features]_Wq
```

42.2 Cosine similarity

```
EET_{j,q} = cosine(StateVec_anchor,j, StateVec_after,q)
= dot(V_j, V_q) / (||V_j|| * ||V_q|| + eps)
```

42.3 Difference from control echo

```
DeltaEET_j = EET_{Target anchor j -> B2} - EET_{Control matched window j -> B2}
```

EET is state-vector resemblance. It is not proof of memory replay.

43. MRED-ITP: Information-Thermodynamic Proxy layer

EXPLORATORY PROXY FAMILY

MRED-ITP tests complexity perturbation/settlement and ocular release around MRED events.

43.1 ACG: Algorithmic Complexity Gate

Given symbolic sequence S_t derived from cleaned EEG window W_t:

```
C_LZ(t) = LZC(S_t) / E[LZC(shuffle(S_t))]
```

For anchor j:

```
C_pre,j = mean(C_LZ over [a_j - 15, a_j])
C_peak,j = mean(C_LZ over [a_j, a_j + 5])
C_post,j = mean(C_LZ over [a_j + 10, a_j + 30])
DeltaC_strike,j = C_peak,j - C_pre,j
DeltaC_settle,j = C_post,j - C_peak,j
DeltaC_baseline,j = C_post,j - C_pre,j
```

ACG flag:

```
ACG_j = 1[DeltaC_strike,j > theta_up]
* 1[DeltaC_settle,j < -theta_down]
* 1[Artifact_j < theta_artifact]
```

43.2 CSI: Complexity Settlement Index

```
CSI_j = z(DeltaC_strike,j) - z(DeltaC_post_minus_peak,j)
```

where DeltaC_post_minus_peak is positive when post remains above peak. A high CSI means perturbation followed by settlement.

43.3 OCU: Ocular-Cognitive Unloading

Blink detection proxy:

```
blink_i = 1[Fp1_p2p_i > theta_p2p
and Fp1_lowfreq_i > theta_lf
and duration_i in [100,500] ms]
```

Blink rates:

```
BR_pre,j = count(blinks in [a_j - 30, a_j - 5]) / 25
BR_hold,j = count(blinks in [a_j - 5, a_j + 5]) / 10
BR_release,j = count(blinks in [a_j + 5, a_j + 20]) / 15
```

Suppression and release:

```
BlinkSuppression_j = z(BR_baseline - BR_hold,j)
BlinkRelease_j = z(BR_release,j - BR_hold,j)
```

OCU index:

```
OCU_j = 0.45*z(BlinkSuppression_j)
+ 0.45*z(BlinkRelease_j)
+ 0.10*z(EventBoundary_j)
- 0.25*z_positive(Artifact_j)
```

Strict OCU flag:

```
OCUFlag_j = 1[BlinkSuppression_j > theta_S]
```

```
* 1[BlinkRelease_j > theta_R]
* 1[Artifact_j < theta_artifact]
```

OCU is not proof of memory encoding.

44. LSO / SPM: Subtitle Override and Subtitle Phase Mapping

EXPLORATORY SUBTITLE CONTROL

This is optional and not part of default PRAYCG2.0 unless selected.

44.1 Subtitle line representation

```
Sub_i = (t_on,i, t_off,i, text_i, bbox_i)
```

44.2 Lexical Extraction Cost

```
CharsPerSec_i = N_chars,i / (t_off,i - t_on,i)
WordsPerSec_i = N_words,i / (t_off,i - t_on,i)
LineLoad_i = N_lines,i
LEC_i = w_c*z(CharsPerSec_i)
+ w_w*z(WordsPerSec_i)
+ w_l*z(LineLoad_i)
+ w_s*z(SegmentationDifficulty_i)
+ w_g*z(GazeRevisits_i)
```

44.3 Subtitle semantic timing

If gaze is available:

```
t_read,i = first time gaze completes subtitle bbox traversal
```

If gaze is unavailable:

```
t_semantic,i = argmax_{t in [t_on,i - 2, t_off,i + 1]} TSP(t)
```

Phase shift relative to audio:

```
phi_i = t_read_or_semantic,i - t_audio_anchor,i
```

44.4 Subtitle MRED

```
MR_sub,j = 0.30*z(MeaningGamma_j)
+ 0.30*z(TSP_j)
+ 0.20*z(KHT_topo_j)
+ 0.20*z(LexicalRecognition_j)
- 0.20*z(Artifact_j)
ENC_sub,j = 0.55*z(H_theta_j)
+ 0.15*z(API_A_j)
+ 0.15*z(Novelty_j)
- 0.20*z(Familiarity_j)
- 0.25*z(LEC_j)
- 0.20*z(TaskGamma_j)
- 0.15*z(Artifact_j)
```

Subtitle choke:

```
SubtitleChoke_j = 1[MR_sub,j > theta_MR]
* 1[ENC_sub,j < theta_ENC]
* 1[LEC_j > theta_LEC]
```

45. Topo-OSM Network State

HUMAN-SCALE TOPOLOGY PROXY

```
Topo-OSM is the human-scale interpretation layer for network-state topology.
```

45.1 Human-scale latent state

```
Y_topo(t) = Phi[
A_theta(t),
A_gamma(t),
A_alpha(t),
TSP(t),
API_A(t),
Resp(t),
Artifact(t),
u(t)
]
```

This is not cellular Y_cell and not K_OSM.

45.2 State vector implementation

```
TopoState(t) = [ThetaTopology(t), GammaTopology(t), AlphaState(t),
TSP(t), NIP(t), API_A(t), Resp(t), Artifact(t)]
```

45.3 Topological shift

For pre/post windows:

```
TopoShift_j = distance( mean(TopoState over W_post,j), mean(TopoState over W_pre,j) )
```

Possible distances:

Euclidean: ||V_post - V_pre||_2 Cosine: 1 - cosine(V_post, V_pre) Mahalanobis: sqrt((V_post - V_pre)^T Sigma^-1 (V_post - V_pre))

45.4 Persistence

```
TopoPersistence_j = mean_{q in post windows}(cosine(V_anchor,j, V_q))
Topo-OSM does not prove memory. It provides a human-scale state-topology proxy.
```

46. Opto-PING / Rung 0 synthetic model

SYNTHETIC IDENTIFIABILITY ONLY

Opto-PING is synthetic identifiability and model-comparison support. It does not prove human biology.

46.1 Orthodox PING skeleton

```
E(t) <-> I(t)
where:
E(t) = excitatory population activity
I(t) = inhibitory population activity
```

46.2 Hidden reciprocal model

```
E_{t+1} = a_E*E_t + eta_E*Y_t + b_E*u_t + eps_E,t
Y_{t+1} = a_Y*Y_t + zeta_E*E_t + b_Y*u_t + eps_Y,t
```

46.3 Loop coefficient

```
K_loop = eta_E * zeta_E
K = sqrt(abs(eta_E * zeta_E))
```

K is more stable than separately interpreting eta and zeta.

46.4 Joint negative log likelihood

For conditions c = 1..C and Kalman innovations v_k^(c) with covariance S_k^(c):

```
NLL(theta) = sum_c sum_k [ log|S_k^(c)| + v_k^(c)^T * inv(S_k^(c)) * v_k^(c) ]
where:
theta = [eta_E, zeta_E, process noise, observation noise, ...]
```

46.5 Model-selection gate

```
Pass_Rung0 = 1[K_error <= threshold]
* 1[FullModel_CV_WinRate >= threshold]
* 1[NullAlternatives do not match full model]
```

Boundary:

Synthetic identifiability only.

No biological or human EEG mechanism claim.

47. CAI/SID v0.2

FROZEN EXPLORATORY CANDIDATE

> Construct boundary
> CAI means Controlled Access-Integration Index. It is a bounded unitless operational proxy, not a probability, diagnosis, direct consciousness measurement, or proof of narrative reception.

47.1 Fast and slow components

```
L_E(t) = weighted_available(.45 z_MeaningGamma, .35 z_TSP, .20 z_NIP)
         - .10 max(z_VisualGamma,0)
E(t) = sigmoid(L_E(t))
L_Y(t) = .50 z_ThetaIntegration + .30 z_NAS + .20 z_PosteriorAlpha
Y(t) = sigmoid(L_Y(t))
```

Available-input means weights renormalize over components present at that row. The source column and adapter grade are recorded. General gamma is not assumed equivalent to MeaningGamma.

47.2 Retrospective future integration

```
Y_future(t) = mean{Y(s): same block, t+8 <= s <= t+30}
```

At least five future samples are required. No branch-end median fill is allowed. Because future samples are used, this is an offline retrospective estimate.

47.3 Corrected positive loop

```
e_plus(t) = max(E(t)-.5,0)
y_plus(t) = max(Y_future(t)-.5,0)
K_product(t) = 2 sqrt(e_plus(t)y_plus(t))
rho_plus(t) = max(corr_local(E,Y_future),0)
K(t) = .70 K_product(t) + .30 rho_plus(t) [if rho estimable]
K(t) = K_product(t) [otherwise]
```

If either E or Y_future is at or below .5, K_product is zero. Jointly low signals can be similar but are not positive-loop evidence. Unestimable correlation never becomes favorable support.

47.4 Artifact, task, and composite

```
A(t)=sigmoid(weighted_available(.55 z_Artifact,.25 z_HF,.20 z_P2P))
X(t)=sigmoid(z_Task)
E_plus=clip01(2(E-.5)); Y_plus=clip01(2(Y-.5))
A_plus=clip01(2(A-.5)); X_plus=clip01(2(X-.5))
Support=mean(E_plus,Y_plus,K)
Penalty=mean(A_plus,X_plus)
CAI_core=Support*(1-Penalty)
```

CAI_core is bounded in [0,1], but boundedness does not imply probability. A hard artifact/HF/P2P violation makes the window missing, not zero.

```
Reliability_context = D_gate*(1-confound_load)*(1-extraction_load)
CAI_context_gated = CAI_core*Reliability_context
Mean_CAI = integral(CAI_core dt)/T_valid
CAI_load = integral(CAI_core dt)
CAI_occupancy(q)=valid_time(CAI_core>=q)/T_valid
SID_[a,b]=(1/(b-a)) integral_a^b CAI_core(t)dt
```

Context gating is a separate attribution sensitivity output. DGA and autonomic outputs are not primary CAI inputs. The exploratory occupancy threshold is .50; moving-block intervals describe within-run temporal stability, not population inference.

48. Micro Handoff v0.1

SOFTWARE-VALID, CONSTRUCT-UNVALIDATED

Boundary

Micro Handoff is not a consciousness detector, probability, diagnosis, proof of discrete 40 Hz frames, or validated consciousness RPM/density gauge.

48.1 Causal feature windows

- 25 ms output update grid, not 25 ms spectral resolution.
- T5/T6 temporal gamma 30-45 Hz with trailing 250 ms Hann periodogram.
- Pz/P3/P4/T5/T6 theta 4-8 Hz with trailing 500 ms Hann periodogram.
- Baseline 1 robust normalization with no full-run fallback.
- Fixed equal-weight lag bank from 100 to 1000 ms.

```
F(t)=sigmoid(z_B1(log temporal-gamma power))
S(t)=sigmoid(z_B1(log theta-integration power))
Q(t)=exp[-.50 max(Z_sentinel(t)-2,0)] * exp[-.15 max(Z_global_p2p(t)-4,0)]
```

Sentinel z > 5, global P2P z > 8, timestamp gaps, nonfinite required values, or rows outside identified blocks are missing.

48.2 Coordination, handoff, and density

```
O(t)=clip01(max(F_100(t)-F_100(t-.1),0)/.25)
R(t,tau)=clip01(max(S_250(t+tau)-S_250(t+tau-.1),0)/.25)
C(t,tau)=1-|F(t)-S(t+tau)|
Q_pair=sqrt(Q(t)Q(t+tau))
H(t,tau)=Q_pair*sqrt(O(t)R(t,tau))
A(t,tau)=[F(t)+S(t+tau)]/2
D(t,tau)=Q_pair*A(t,tau)*C(t,tau)
```

H and D are equal-weight means across the full lag bank. Low-low can have high agreement C but low H and D. Stationary high-high can have high D but low H. Directional H requires a positive fast rise followed by a positive slow rise.

48.3 Validation status

All 14 declared synthetic behavior checks passed. A planted 500 ms raw-signal handoff was recovered at 600 ms, within the declared +/-100 ms tolerance caused by unequal trailing spectral windows. Synthetic success validates software behavior only.

Historical Target-Control metric	Mean	Positive runs	Run-bootstrap interval
Positive directional handoff	+0.00056	3/6	[-0.0016,+0.0026]
Activation-weighted density	+0.0441	5/6	[+0.0050,+0.0926]
Coordination	+0.00127	3/6	[-0.0228,+0.0305]

Twenty-four unique historical XDF files were inventoried; six were eligible with caution at 125 Hz. Directional handoff was mixed, diagnostic lags varied, and no run passed both time-structure null families.

PART VII
Expanded confound defenses

Challenges sensory, spatial-attention, order, afterglow, and respiratory explanations before interpretation.

49. HOC-R: high-order confound residualization

ATTRIBUTION AUDIT

```
y = X beta + error
R2_train = 1 - sum(y-y_hat)^2/sum(y-mean(y))^2
```

HOC-R inventories low-level visual/audio, cue, and high-order face/body/object/AOI regressors and reports whether residualization is supportable. Missing high-order regressors leave Target-Phase cautioned.

50. OSA: ocular and spatial-attention burden

EXPLORATORY CONFOUND DIAGNOSTIC

```
cue_duty = display_duration/cue_interval
report_burden = mean_available(legibility,blur,eye_strain,stall,task_load)
OSA = mean_available(cue_duty,report_burden)
```

The current caution flag uses OSA >= .35 when gaze/AOI data are absent. The threshold is exploratory.

51. OHC: order, habituation, and carryover

WARNING HEURISTIC

OHC records branch order and refuses to treat a 120-second washout as a full biological reset. Risk values are warnings, not probabilities. Counterbalancing reduces systematic imbalance but does not eliminate carryover.

52. AAM: afterglow attribution model

EXPLORATORY ATTRIBUTION DIAGNOSTIC

```
sim(x,y)=dot(x,y)/(||x||||y||)
NAI=mean_available(sim(B2,W2),target_echo,target_afterglow,story_active_W2)
TCRI=mean_available(task_load,compliance,stall,sim(B2,W3))
AAM=NAI-TCRI
```

AAM > .20 is Target-afterglow-like, AAM < -.20 is task-relief-like, and the middle is mixed/ambiguous under the current heuristic. Baseline 2 remains cumulative, not Target-only.

53. RespDualPath: legacy and continuous respiratory paths

RESPIRATORY ALTERNATIVE EXPLANATION

```
d_t = |resp_t-resp_(t-1)|
threshold = mean(d)+2.5 SD(d)
```

Legacy feature-table events above threshold are preserved as possible state events and cautioned when artifact overlaps. The continuous v1.0 path is preferred when raw respiratory and cardiac streams are available.

PART VIII
Software workflow, inference, and interpretation

Defines what software may conclude, how status travels through chained modules, and what must be frozen prospectively.

54. Visualizer formulas

AUDIT AND SYNCHRONIZATION DISPLAY

The visualizer does not create scientific endpoints. It audits synchronization and module outputs.

54.1 Robust axis scaling

For visible panel window W_vis:

```
y_min = percentile(y over W_vis, 5)
y_max = percentile(y over W_vis, 95)
```

or z-score mode:

```
y_plot = clip(rz(y), -z_clip, z_clip)
```

54.2 Missing-signal gate

```
UsableFeature = 1[non_null_count >= N_min]
* 1[std(y) > eps]
* 1[unique_count > 1]
```

If UsableFeature = 0, the panel should display "signal unavailable" rather than a fake flat line.

54.3 Event card display

For event time e_k:

```
ShowEventCard_k(t) = 1[ |t - e_k| < card_window ]
```

55. Offline interpretive report generator

DETERMINISTIC EXPLANATION ONLY

The offline interpreter is deterministic and rule-based.

55.1 Table detection

```
ModulePresent_m = 1[expected_table_m exists]
```

55.2 Rule evaluation

Example:

```
if A_MRED_pass_count > 0 and claim_level strict:
statement = "A-MRED positive under current strict gate."
elif A_MRED_pass_count > 0 and timing_caution:
statement = "A-MRED pilot-positive with timing caution."
```

```
else:
statement = "No strict A-MRED pass detected."
```

55.3 Boundary

The offline interpreter explains tables.

It does not create new evidence.

It does not replace expert review.

56. Canonical frame and status-aware chaining

IMPLEMENTED ANALYSIS CONTRACT

Every time-resolved row identifies source time, canonical condition, unique contiguous block instance, source columns, adapter grade, evidence grade, hard-QC state, and missingness. Repeated occurrences of one condition cannot share future windows across block boundaries.

Status	Downstream meaning
PASS	Declared requirements satisfied.
CAUTION	Result may be computed, but warnings propagate.
INELIGIBLE	Endpoint cannot be interpreted.
NOT_APPLICABLE	Protocol lacks required condition/input.
FAILED	Execution failed; absence cannot become evidence.

Exploratory troubleshooting output after a failed primary gate cannot inherit primary or confirmatory status and cannot alter Master Suite eligibility.

57. Module interaction map

PROCESSING MAP

A single event can pass through the suite as:

1. MediaPrep creates stimulus + cue + anchor scaffold.
2. Runner records which files were used and writes Stasis markers.
3. Feature extractor computes EEG/autonomic/stimulus features.
4. ArtifactScore and confound registry define veto/caution layers.
5. TSP/GammaScalpel detect candidate recognition timing.
6. Theta/Topo/after-state features estimate integration.
7. MRED separates recognition from integration.
8. A-MRED applies the strict primary endpoint gate.
9. NIP/CII summarizes continuous immersion density.
10. TTI compares reception vs extraction.
11. NUPI classifies update polarity.
12. CET-R tests whether stimulus rhythm explains the effect.
13. EET/MRED-ITP/OCU/ACG provide exploratory convergence.
14. Visualizer and offline interpreter explain the result without adding evidence.

58. Example event derivation

WORKED COMPUTATIONAL PATH

For a predeclared Target anchor j:

Step 1: compute MeaningGamma_j and TSP_j.

Step 2: compute MR_T,j.

Step 3: compute H_theta_10_30,j, TopoShift_j, API_A_post,j.

Step 4: compute ENC_T,j.

Step 5: repeat matched windows for Control and Override.

Step 6: compute ArtifactPass, ConfoundPass, TimingPass, CETRPass.

Step 7: compute A_MRED_j.

Step 8: compute CII_T/C/O and IAQ.

Step 9: compute PeakEvidence and ResolutionEvidence.

Step 10: write visual overlay and offline interpretation.

Mathematically:

```
A_MRED_j = Gate(MR_T,j, ENC_T,j, MR_C,j, ENC_C,j, MR_O,j, ENC_O,j, QC_j)
CII_margin_j = CII_T,j - max(CII_C,j, CII_O,j)
PeakEvidence_j = f(MR_T,j, ENC_T,j, CII_margin_j, KHT_topo_j, QC_j)
ResolutionEvidence_j = f(B2_echo_j, EET_j, RDI_j, self_report_echo_j, API_recovery_j)
```

59. Prospective inferential model

PRE-REGISTRATION CANDIDATE

```
Y_(participant,stimulus,condition) = beta0 + beta_condition + period
                                    + u_participant + v_stimulus + error
u_participant ~ N(0,sigma_u^2)
v_stimulus ~ N(0,sigma_v^2)

PRAYCG4 contrasts:
Target - PhaseScrambled
Target - ShotOrderScramble
Target - ContextualOverride
```

Multiplicity, exclusions, coverage thresholds, lag windows, nulls, random effects, and carryover terms must be frozen before confirmatory outcome review. Independent recognition, comprehension, confidence, and narrative-coherence outcomes are criterion evidence, not CAI ingredients.

60. Confirmation-grade decision rule

REQUIRED AUDIT CHECKLIST

A run should be called confirmation-grade only if all of the following are true:

```
FrameVerifiedLockedAnchorFileLoaded = TRUE
AnchorHashRecordedByRunner = TRUE
ALS_Pulse_PASS_Control_Target_Override = TRUE
TargetOverrideHashMatch = TRUE
ControlGeneratedFromCueEmbeddedTarget = TRUE
ControlAudioQC = PASS
StimulusFingerprintAvailable = TRUE
CET_R_DoesNotExplainEndpoint = TRUE
ArtifactGate = PASS
ConfoundGate = PASS
A_MRED_or_MREDPeakResolution = PASS
SelfReportNotContradictory = TRUE
```

If any key component fails, label the run using the correct caution:

estimated-anchor pilot operator-frame-verified but ALS-cautioned runner-registered but audio-confounded conceptually predeclared / not runner-registered PRAYCG-lite / no sensory-control branch The point is not to make confirmation impossible. The point is to make confirmation auditable.

61. Promotion and falsification

PROSPECTIVE PROMOTION GATES

- Software equations, configuration, code, and reports agree within declared tolerance.
- Time-shift, phase-randomized, pseudo-event, and sensitivity nulls behave as preregistered.
- Results do not depend on one optional component, stimulus, or post-hoc lag.
- The frozen candidate predicts independent criteria in new participants and stimuli.
- Reliability, uncertainty, held-out performance, and any probability calibration are established.
- Null, reversed, and ineligible results remain visible; no post-hoc archetype rescues them.

62. Formula safety notes

REQUIRED LANGUAGE DISCIPLINE

62.1 The suite contains operational indices

These formulas are engineered proxies. They are not direct measurements of hidden essence.

```
MeaningGamma != meaning
TSP != true semantic time
KHT-topo != K_OSM
EET != replay proof
OCU != memory dump proof
NUPI != heat or ATP
TTI != moral theft
A-MRED != consciousness proof
```

62.2 Positive result language

Use:

Target-specific recognition-plus-integration candidate.

A-MRED pass under current gates.

MRED-Peak candidate.

MRED-Resolution candidate.

Resolvable recovery profile.

Accommodative-load profile.

Avoid:

proved consciousness proved memory formation proved OSM biology proved the soul proved thermodynamic meaning

62.3 Negative result language

Use:

No strict pass under current operational criteria.

Recognition may have occurred without detectable integration.

Timing or artifact caveats prevent strict interpretation.

Avoid:

nothing happened the scene was meaningless the subject felt nothing

63. Minimal confirmatory endpoint recommendation

PROSPECTIVE PRIMARY ENDPOINT GUIDANCE

For a future group or preregistered study, the primary endpoint should be one of:

A-MRED pass proportion across locked anchors

```
DeltaMRED continuous score across locked anchors
```

MRED-Peak vs MRED-Resolution classification across locked anchors The confirmatory statistical model could be:

```
A_MRED_pass ~ Condition + (1|Subject) + (1|Stimulus) + (1|Anchor)
```

or continuous:

```
DeltaMRED ~ Condition + Familiarity + Artifact + CETR + (1|Subject) + (1|Stimulus) + (1|
Anchor)
```

Prediction:

```
Target > Control
Target > Override
```

The rest of the suite should be treated as secondary or exploratory unless preregistered as primary.

64. Public claim boundary

REQUIRED PUBLIC WORDING

Permitted	Not permitted
CAI v0.2 was higher/lower under the frozen model.	CAI measures or proves consciousness.
Micro Handoff did/did not exceed declared nulls.	Gamma-to-theta handoff proves 40 Hz conscious frames.
ShotOrder tests temporal structure beyond local shots.	Target-ShotOrder uniquely isolates meaning.
Autonomic arrays estimate timing and respiratory dependence.	HRV increase proves meaning or parasympathetic truth.
A hash verifies file identity.	A hash proves scientific validity.

PART IX
Outputs, provenance, and reference catalog

Lists expected products, file ownership, compact formulas, and methodological references.

65. Recommended output files by module

OUTPUT CONTRACT

Core feature extraction:

*_time_resolved_feature_frame.csv *_phase_feature_summary.csv A-MRED:

amred_anchor_endpoint_table.csv amred_primary_endpoint_summary.csv amred_visual_overlay.csv MRED-Peak/Resolution:

mred_peak_resolution_anchor_table.csv mred_peak_resolution_run_summary.csv top_mred_peak_candidates.csv top_mred_resolution_candidates.csv mred_peak_resolution_visual_overlay.csv MRED / KHT:

mred_event_table.csv candidate_local_kht_topo_mred_event_table.csv candidate_local_kht_visual_overlay.csv NIP / CET / EET:

nip_component_timeseries.csv bit_event_table.csv cii_anchor_integrals.csv iaq_target_override_table.csv cet_residualization_model_summary.csv eet_endogenous_echo_tracking.csv TTI / NUPI:

tti_global_summary.csv tti_anchor_deltas.csv nupi_run_summary.csv nupi_anchor_polarity_table.csv OCM/RSM/CVB/Squint:

ocm025_rsm_cvb_squint_cue_epoch_table.csv rsm_correlation_summary.csv confound_registry.csv MRED-ITP:

lz_complexity_timeseries.csv acg_event_table.csv blink_event_table.csv ocu_event_table.csv mred_itp_anchor_summary.csv Visualizer:

*_features_used.csv *_events_used.csv *_feature_diagnostics.csv *_event_time_diagnostics.csv *_render_report.json Offline interpreter:

reports/offline_interpretation/offline_interpretive_report.md reports/offline_interpretation/offline_interpretive_report.txt reports/offline_interpretation/offline_interpretive_report_summary.json

66. File provenance rule

OWNERSHIP AND TRACEABILITY

MediaPrep:

stimulus videos, cue schedules, draft/locked anchor scaffolds, stimulus fingerprints.

Protocol runner:

event markers, media file hashes, run configuration, self-reports, confound reports.

Master Comprehensive Suite:

feature tables, module outputs, event tables, endpoint summaries.

Visualizer:

synchronized MP4 and sidecar tables.

Offline interpreter:

deterministic text interpretation of existing module outputs.

Clean rule:

If a file describes what was shown, it probably comes from MediaPrep or the runner.

If a file describes physiology over time, it comes from the Master Comprehensive Suite.

If a file explains what the outputs mean, it comes from the offline interpreter or report generator.

67. Compact formula catalog

QUICK-REFERENCE EQUATIONS

```
rz(x) = (x - median_ref) / (1.4826*MAD_ref + eps)
PLV = |mean(exp(i*(phase_a - phase_b)))|
ArtifactScore = mean(rz(p2p), rz(HF_45_55), rz(EMG), rz(FpSentinel), rz(line_noise),
step_flag)
API_A = .30*z(-HR) + .25*z(RMSSD) + .20*z(SDNN) + .15*z(pNN50) + .10*z(-HR_slope) -
.10*z(RespInstability)
TSP = w_G*MeaningGamma + w_PT*PosteriorTemporal + w_N*NAST + w_K*KHT -
w_X*TaskGamma - w_A*Artifact - w_V*VisualDrive
KHT_topo = sqrt(abs(eta_HT*zeta_HT))
MR = w_TSP*TSP + w_G*MeaningGamma + w_K*KHT + w_N*NAST - penalties
ENC = w_H*ThetaHandoff + w_T*TopoShift + w_E*EET + w_API*API_A + w_B2*B2Echo -
penalties
A_MRED = 1[MR_T>th] * 1[ENC_T>th] * 1[T>C] * 1[T>O] * 1[QC=PASS]
NIP_density = clip_plus(A_sem)^alpha * clip_plus(R_int_lag)^beta * Penalty
CII = mean(NIP_density over anchor window)
IAQ = 1 - (CII_Override + eps)/(CII_Target + eps)
TTI = w1*z(MR_T-MR_O)+w2*z(ENC_T-ENC_O)+w3*z(API_T-API_O)+w4*z(X_O-X_T)-
penalties
NUPI = RDI - ALI
CET residual: Y_resid = Y - (beta0 + beta_u^T U + beta_art*Artifact)
EET = cosine(StateVec_anchor, StateVec_after)
LZC_norm = LZC(S)/E[LZC(shuffle(S))]
OCU = .45*z(BlinkSuppression)+.45*z(BlinkRelease)+.10*z(EventBoundary)-.25*z(Artifact)
LEC = w_c*z(chars/sec)+w_w*z(words/sec)+w_l*z(line_count)+w_s*z(segmentation)
+w_g*z(gaze_revisits)
OptoPING K = sqrt(abs(eta_E*zeta_E))
```

68. Current additions to the compact catalog

QUICK-REFERENCE EQUATIONS

```
ShotOrder = concatenate(S_pi(1),...,S_pi(m))
SZI = product(1-Meaning,1-Empathy,1-Absorption,1-Threat,1-PuzzleDemand)
K_CAI = .70[2sqrt((E-.5)_+(Yfuture-.5)_+)] + .30(corr)_+
CAI_core = mean(E_plus,Y_plus,K)[1-mean(A_plus,X_plus)]
RMSSD = sqrt(mean(diff(RR)^2))
SDNN = sample_sd(RR); pNN50 = mean(|diff(RR)|>50ms)
p_shift = [1+count(|null|>=|observed|)]/(B+1)
H_micro = sqrt(Q_t Q_t+lag)sqrt(positive_fast_change*positive_slow_change)
D_micro = sqrt(Q_t Q_t+lag)mean(F,Sfuture)[1-|F-Sfuture|]
AAM = NAI-TCRI
```

69. Methodological references

SELECTED FOUNDATIONS

- Task Force of the European Society of Cardiology and NASPE (1996). Heart rate variability: standards of measurement, physiological interpretation and clinical use. Circulation 93:1043-1065. PMID 8598068.

- Williams, E. J. (1949). Experimental Designs Balanced for the Estimation of Residual Effects of Treatments. Australian Journal of Scientific Research, Series A 2:149-168. DOI 10.1071/CH9490149.
- Stojanoski, B., and Cusack, R. (2014). Time to wave good-bye to phase scrambling. Journal of Vision 14(12). DOI 10.1167/14.12.6.
- Yuval-Greenberg, S. et al. (2008). Transient induced gamma-band response in EEG as a manifestation of miniature saccades. Neuron 58(3):429-441. PMID 18466752.

70. Final statement

GOVERNING CONCLUSION

> Layered falsification system
> The strongest contribution of PRAYCG is not that every module yields a positive number. It is that timing, artifact, sensory structure, local shot content, temporal order, task stance, gaze burden, respiration, autonomic quality, carryover, self-report, and independent criteria can constrain the same candidate result under declared rules.

Every formula in this report has a mathematical address, an evidence grade, and a claim boundary. These are auditable operational definitions, not metaphysical declarations.